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Predicting the Quality of Life Index: A Comparative XGBoost–LSTM Study with SHAP-Based Explainable AI

İbrahim Budak ¹

Abstract

In this study, the prediction performance of different artificial intelligence algorithms was examined using quality of life data from 2016 to 2025. The analysis compared gradient-boosted tree-based XGBoost with LSTM, which has the capacity to model time series and sequential dependencies. In addition, SHAP analysis was applied to ensure the model's explainability and to identify the key factors affecting quality of life. The findings show that both models successfully capture quality of life patterns, with the LSTM model achieving higher out-of-sample accuracy than XGBoost (higher R^2 and lower MAE, RMSE, and MAPE). SHAP analysis revealed that Purchasing Power and Pollution are the factors with the strongest impact on quality of life. The decisive effect of Purchasing Power indicates that macroeconomic conditions such as real income level, price stability, and Purchasing Power Parity - adjusted welfare indicators directly reflect quality of life. Other factors, such as cost of living, housing price/income ratio, security, healthcare services, climate, and commute time, were found to have varying degrees of importance across countries. These findings emphasize the priority of designing macroeconomic frameworks targeting income/wage policies and price stability alongside policies aimed at improving environmental conditions. The results obtained indicate that policy makers should focus on the efficient allocation of resources. The results obtained provide policymakers with an evidence-based roadmap for the efficient allocation of resources and demonstrate that more detailed analyses can be conducted using different explainable artificial intelligence methods for future research. Additionally, to test the robustness of the model, different training/testing splits, alternative error metrics, and hyperparameter sensitivity analyses were performed; the direction and magnitude of the main findings were found to be consistent across these scenarios. Finally, SHAP-based findings provide a starting framework for policy simulations, enabling the quantitative prediction of potential welfare gains from targeted improvements in specific sub-indices.

Keywords: Quality of Life, Purchasing Power, XGBoost, LSTM, SHAP, Explainable AI

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2025, 14 (5), 2208-2224 | Araştırma Makalesi

Yaşam Kalitesi Endeksinin Tahmini: SHAP Tabanlı Açıklanabilir Yapay Zeka ile Karşılaştırmalı Bir XGBoost–LSTM Çalışması

İbrahim Budak¹

Öz

Bu çalışmada, 2016–2025 yılları arasındaki yaşam kalitesi verileri kullanılarak farklı yapay zekâ algoritmalarının tahmin performansları incelenmiştir. Analizde, gradyan artırılmalı ağaç tabanlı XGBoost ile zaman serisi ve ardışık bağımlılıkları modelleme kapasitesine sahip LSTM karşılaştırılmıştır. Ayrıca, modelin açıklanabilirliğini sağlamak ve yaşam kalitesini etkileyen temel faktörleri belirlemek amacıyla SHAP analizi uygulanmıştır. Bulgular, her iki modelin de yaşam kalitesi eğilimlerini başarılı şekilde yakaladığını, ancak LSTM'in daha yüksek doğruluk ($R^2=0.98$) ve daha düşük hata değerleri (MAE, RMSE, MAPE) ile XGBoost'a göre üstün performans sergilediğini göstermektedir. SHAP analizi, yaşam kalitesi üzerinde en güçlü etkiye sahip faktörlerin Satın Alma Gücü ve Kirlilik olduğunu ortaya koymuştur. Satın Alma Gücü'nün belirleyici etkisi, reel gelir düzeyi, fiyat istikrarı ve satın alma gücü paritesi–düzeltilmiş refah göstergeleri gibi makroekonomik koşulların yaşam kalitesine doğrudan yansıdığını göstermektedir. Yaşam maliyeti, konut fiyat/gelir oranı, güvenlik, sağlık hizmetleri, iklim ve trafik süresi gibi diğer faktörlerin ise ülkeler arasında değişen önem derecelerine sahip olduğu belirlenmiştir. Bu bulgular, gelir/ücret politikaları ve fiyat istikrarını hedefleyen makroekonomik çerçeveler ile çevresel koşulların iyileştirilmesine yönelik politikaların birlikte tasarlanmasının öncelikli olması gerektiğini vurgulamaktadır. Elde edilen sonuçlar, politika yapımcılar için kaynakların verimli yönlendirilmesine yönelik kanıta dayalı bir yol haritası sunmakta ve gelecekteki araştırmalar için farklı açıklanabilir yapay zekâ yöntemleri ile daha ayrıntılı analizlerin yapılabileceğini göstermektedir. Ek olarak, model sağlamlığını sınamak amacıyla farklı eğitim/test bölmeleri, alternatif hata metrikleri ve hiperparametre duyarlılık analizleri gerçekleştirilmiş; ana bulguların yönü ve büyüklüğü bu senaryolarda da tutarlı bulunmuştur. Son olarak, SHAP tabanlı bulgular politika simülasyonları için bir başlangıç çerçevesi sağlayarak, belirli alt endekslerde hedeflenen iyileştirmelerin olası refah kazanımlarını nicel olarak öngörmeye imkân tanımaktadır.

Anahtar Kelimeler: Yaşam Kalitesi, Satın Alma Gücü, XGBoost, LSTM, SHAP, Açıklanabilir Yapay Zekâ

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Introduction

The concept of “quality of life (QoL)” is becoming increasingly central to understanding the welfare levels of societies. Economic development, accessibility of healthcare services, environmental factors, security, and social welfare indicators are among the fundamental elements that shape the quality of life for individuals and societies (Cummins, 1996, p.304; Diener et al., 2018, p.253). Therefore, measuring quality of life plays a critical role not only in sociology and health sciences, but also in economics, urban planning, and policy development.

Quality of life indices have emerged as a product of efforts to express this multidimensional concept numerically. Tools such as Numbeo, the OECD Better Life Index, and the World Health Organization's quality of life scales assess the welfare levels of societies in a comparable manner, taking into account sub-dimensions such as purchasing power, environmental pollution, access to healthcare, housing costs, safety, transportation, and climate (Numbeo, 2025; WHOQOL Group, 1998, p.551). These sub-indices guide policymakers and researchers in making comparisons between countries and cities. Similarly, Alkaya (2024) analyzes the positions of OECD countries based on the Better Life Index using multidimensional scaling, highlighting the multidimensional nature of well-being and cross-country disparities in quality of life (Alkaya, 2024, p.1130).

Although quality of life indices are widely used in comparative welfare research, a substantial part of the empirical literature still relies on classical statistical models, such as multiple linear regression or panel data approaches, which often assume linear relationships and limited interaction structures between variables (Rahman et al., 2005; Bingöl & Ayhan, 2020, p. 441–468; Rizzo, 2019, p.3). In addition, many studies either focus on a single cross-section or a short time span, which restricts the analysis of long-term dynamics in quality of life indicators (Abegaz et al., 2025, p.166).

From a methodological perspective, the prediction of QoL is increasingly carried out with machine learning algorithms, including tree-based ensembles and deep neural networks. Recent studies have applied XGBoost, LSTM and related models to health-related QoL scores, urban quality-of-life indicators and composite welfare indices, often in combination with explainable AI tools such as SHAP to identify key predictors and their directions of effect. Nevertheless, there is still relatively limited work that jointly (i) compares tree-based and sequential architectures on a harmonized multi-country QoL panel derived from Numbeo, and (ii) links SHAP-based explanations directly to macroeconomic and environmental policy channels in a transparent way (Jung & Yun, 2025, p.1; Zhu et al., 2024, p.9551; Panat & Chandra, 2025, p.1; Shu et al., 2022, p.1).

The rapid advancement of artificial intelligence and machine learning techniques in recent years has opened up new possibilities for predicting complex and multi-variable social indicators such as quality of life. Tree-based methods (e.g., XGBoost) are preferred for their high accuracy and interpretability, while deep learning methods (e.g., LSTM) stand out for their potential to model the time dimension and dynamic relationships between variables (Chen & Guestrin, 2016, p.1; Hochreiter & Schmidhuber, 1997, p.1735-1736). However, understanding the decision mechanisms of machine learning models is also of great importance. At this point, the SHAP (SHapley Additive exPlanations) method is used as a powerful tool that explains which factors are decisive in the predictions of models (Lundberg & Lee, 2017, p.3).

Despite the availability of composite QoL indices, an open empirical question remains as to which of the underlying sub-indices (purchasing power, pollution, housing affordability, cost of living, safety, health, climate and commute time) are most responsible for cross-country and intertemporal variation in the overall index once all components are considered jointly. From a modelling perspective, it is also not clear *ex ante* whether tree-based ensemble methods or sequential neural-network architectures are better suited to reproducing the composite QoL index from its components under realistic data constraints.

Against this background, this study aims to compare the performance of XGBoost and LSTM algorithms in predicting the Numbeo composite Quality of Life Index from its eight published sub-indices using an annual country–year panel between 2016 and 2025 and to examine the key factors determining quality of life through SHAP analysis. Thus, both the strengths and weaknesses of the methods in terms of prediction accuracy will be evaluated, and the weights of sub-indices such as purchasing power, environmental pollution, security, and health services on quality of life will be revealed.

This study contributes to the growing literature at the intersection of QoL measurement and machine learning in three ways. First, by jointly evaluating XGBoost and LSTM on a harmonized quality of life dataset covering 88 countries for the 2016–2025 period, it provides empirical evidence on the added value of exploiting temporal dynamics compared to a strong tree-based ensemble benchmark on tabular data. Second, the study integrates SHAP-based explainability into QoL prediction and quantifies the relative importance and direction of eight key sub-indices (purchasing power, pollution, housing price/income ratio, cost of living, safety, health care, climate, and commute time) within a single, transparent framework. Third, by linking the SHAP results to concrete policy channels and conducting robustness checks with alternative train–test splits and hyperparameter settings, the analysis offers a policy-oriented and methodologically robust roadmap for decision-makers who seek to prioritize interventions that most effectively improve quality of life.

The remaining sections of the article are structured as follows: The second section presents the quality of life literature, the data sets used, and the methodological approaches in detail. The third section compares the prediction performance of the XGBoost and LSTM algorithms and reports the findings of the SHAP analysis. The fourth section discusses the results and relates the findings to the literature. Finally, the fifth section addresses the limitations of the research and suggestions for future studies.

Literature Review

Quality of life (QoL) is a multidimensional construct influenced by a combination of economic, social, environmental, and health-related indicators. While early approaches to measuring the concept relied on subjective well-being and life satisfaction scales (Diener, Oishi & Tay, 2018, p.253-254), index-based frameworks enriched with institutional indicators (WHOQOL Group, 1998, p.551; OECD, 2013) have become widespread in recent years. In these frameworks, country/city comparisons are made using sub-dimensions such as purchasing power, security, health services, cost of living, housing accessibility, environmental pollution, traffic, and climate. With the proliferation of machine learning, a prediction literature has emerged that goes beyond classical statistical modeling (e.g., multiple linear regression, panel data models) of QoL. Tree-

based methods (random forests, gradient boosting, XGBoost) are particularly preferred due to their sensitivity to non-linear relationships and interactions between variables (Chen & Guestrin, 2016, p.785-786). These methods have been used to predict the relationships between QoL and economic prosperity, environmental quality, and public safety in different regions; model performance has often been shown to be higher compared to classical regression. For datasets with a time dimension, recurrent networks (RNNs) such as LSTM, which are strong at capturing sequential dependencies, have come to the fore (Hochreiter & Schmidhuber, 1997, p.1735-1737). This approach is particularly capable of modeling trends and lagged effects in multi-year index series such as the 2016–2025 country–year QoL panel used in this study.

High accuracy alone is not sufficient; the explainability of decisions is a critical requirement for policy and public research. In this regard, the SHAP (SHapley Additive exPlanations) method calculates the additive contribution of each feature to the individual prediction based on Shapley values in game theory and provides global/local explanations (Lundberg & Lee, 2017). The main advantages of SHAP are that local contributions can be aggregated in a model-agnostic framework, the sign and magnitude are interpretable, and fast inference algorithms exist for tree-based models. Alternative XAI approaches include LIME (local linear approximation), post-hoc explanations using rule extraction and decision trees, visualizations based on partial dependence plots (PDP) and additive local effects (ALE) graphs, and global importance measures (gain/cover/weight) (Ribeiro, Singh & Guestrin, 2016, p.1135-1144; Molnar, 2022, p.121; Arrieta et al., 2020; Doshi-Velez & Kim, 2017). However, given LIME's sampling sensitivity and stability issues, and the risk that PDP/ALE may not fully reflect interactions, SHAP stands out as a more consistent and easily transferable option to decision-makers in most scenarios.

Machine learning adds value in QoL analysis in three dimensions. First, tree-based ensemble methods (especially XGBoost) reduce error metrics by capturing complex, nonlinear relationships and interactions; they are suitable for jointly modeling sub-indices with multicollinearity and scale differences (Chen & Guestrin, 2016, p.785-794). Second, sequential models such as LSTM can learn dependencies, trends, and possible delayed effects between consecutive years; thus, temporal patterns such as the impact of specific policy changes are better captured (Hochreiter & Schmidhuber, 1997, p.1735-1780). Third, feature contributions can be presented at both the global (average effect) and local (country/city-level) levels with SHAP; this approach enables the comparison of relative weights of purchasing power and pollution, the reporting of “equivalent benefit trade-offs,” and scenario analysis (how much does QoL change if a specific sub-index increases?) (Lundberg & Lee, 2017; Molnar, 2022, p.128). Within this integrated framework, establishing XGBoost as a robust baseline model for tabular data, testing LSTM to examine memory effects across years, and using SHAP to make the decision mechanism transparent are consistent with best practices recommended in the literature (Arrieta et al., 2020; Doshi-Velez & Kim, 2017).

Method and Methodology

Data Set and Features

The dataset used in this study was obtained from the Numbeo Quality of Life indices for the period 2016–2025. For each country and year, Numbeo reports eight sub-indices that enter the composite QoL index (purchasing power, pollution, house price to income ratio,

cost of living, safety, health care, traffic commute time and climate). In the modelling stage, we use the annual country–year panel (up to ten observations per country) so that temporal dynamics and lagged effects in QoL can be captured. In addition, for descriptive purposes we compute ten-year averages of the indices for each country; these averages are only used to summarize cross-country differences (Table 1) and do not replace the underlying annual panel used by the prediction models. The source of the dataset is explained in detail in Numbeo's methodology document (Numbeo, 2025).

The quality of life index used includes the following eight sub-dimensions:

- Purchasing Power Index: Represents purchasing power calculated by the ratio of local income to price levels. A high value indicates a better standard of living.
- Pollution Index: It reflects perceptions of air, water, and environmental pollution. A lower value indicates better environmental quality.
- House Price to Income Ratio: It is the ratio of average housing prices to household income. A lower value indicates more affordable housing conditions.
- Cost of Living Index: It measures the price level of consumer goods and services. A lower value indicates a lower cost of living.
- Safety Index: Suç oranları ve bireylerin güvenlik algısını yansıtır. Yüksek değer daha güvenli ortamı gösterir.
- Health Care Index: It measures the accessibility, quality, and satisfaction level of the healthcare system. A high value indicates better healthcare services.
- Traffic Commute Time Index: It shows the average daily commute time for individuals. A lower value reflects less traffic congestion.
- Climate Index: It assesses the impact of climate on quality of life based on temperature, humidity, and meteorological conditions. A higher value indicates a more favorable climate.

These eight sub-indices have different weights in Numbeo's quality of life formula and produce a combined “Quality of Life Index” value. In our study, this composite index was also used as the target variable (dependent variable) of the model.

Using the same sub-indices both as predictors and as components of the target variable raises the possibility of so-called “target leakage”: very high predictive performance may partly reflect the model's ability to reconstruct Numbeo's aggregation rule rather than to predict an external welfare outcome independent of the index definition. In this study, our explicit objective is to approximate the published Numbeo Quality of Life Index from its constituent sub-indices and to provide a transparent decomposition of the composite index into its main drivers. Therefore, we interpret the models as tools for analysing the internal structure of the index rather than as predictors of an external concept of well-being. The implications of this definitional relationship and possible extensions using independent outcome variables (e.g. life satisfaction scores, life expectancy, GDP per capita) are revisited in the limitations and future-research sections.

Furthermore, in the dataset, indices defined as “lower value is better” (e.g., Pollution Index, House Price to Income Ratio, Cost of Living Index, Traffic Commute Time Index)

have been appropriately reversed during the analysis process and made comparable with other indices.

Since the number of countries reporting varies each year, only countries existing in 2025 were used to ensure consistency in the analysis, and index values from previous years were filtered according to this list of countries. Thus, comparisons were made within the same sample framework throughout the entire period.

Table 1. Descriptive statistics of quality of life indices for the period 2016–2025

Variable	N	Mean	Std. Dev.	Min	Max	Direction
Quality of Life Index	88	134.988	35.637	44.533	198.475	Higher is better
Purchasing Power Index	88	65.177	32.107	10.25	139.775	Higher is better
Pollution Index	88	53.768	20.681	12.65	88.462	Lower is better
House Price to Income Ratio	88	13.42	7.422	2.88	53.14	Lower is better
Cost of Living Index	88	50.603	19.123	21.72	120	Lower is better
Safety Index	88	59.711	13.788	17.32	85.633	Higher is better
Health Care Index	88	64.144	10.106	38.58	85.162	Higher is better
Traffic Commute Time Index	88	36.024	8.272	20.5	62.8	Lower is better
Climate Index	88	74.944	18.512	20.2	99.8	Higher is better

Table 1 reports descriptive statistics for the country-level ten-year averages (2016–2025). While these aggregates are useful to highlight structural cross-country differences, the prediction models are estimated on the underlying annual country–year observations. As shown in Table 1, the average and distribution values of the indices show significant differences between countries. For example, the Purchasing Power and Healthcare indices exhibit high variance, while the Climate and Security indices show a relatively narrower distribution. Furthermore, indices based on the “lower value is better” logic (Pollution, Housing Price/Income, Cost of Living, and Travel Time) are presented in the table with their original values and have been appropriately reversed during the analysis process. This information is important for the correct interpretation of the direction and effects of the variables during the modeling phase.

Data Preprocessing

Before proceeding to the modeling phase on the dataset, preprocessing steps were applied. First, missing data analysis was performed. In Numbeo data, index values are not available for some years or some countries. To ensure the integrity of the panel, we restricted the sample to countries that reported QoL indices in 2025 and then collected annual index values for these countries over the 2016–2025 period. This yields an unbalanced panel of country–year observations with up to ten years per country and ensures that the set of countries is constant over time. When missing values were observed, the relevant observations were excluded to avoid bias in the modeling phase. Due to the structure of the data set, missing data were limited, so advanced filling methods such as multiple imputation were not required.

In the second stage, data transformation was performed taking into account the directions of the indices. High values in the Purchasing Power, Security, Health Services, and Climate indices indicate a better quality of life, while low values in the Pollution, Housing

Price/Income Ratio, Cost of Living, and Travel Time indices represent more favorable conditions. Therefore, to increase comparability in the analysis, the indices where “lower values are better” were appropriately reversed, and all variables were adjusted to have the same direction (higher values are better).

In the third stage, normalization was applied when deemed necessary for modeling. Specifically, z-score normalization was preferred to ensure balance in machine learning algorithms for indices measured on different scales (e.g., Health Services on a 0–100 range and Purchasing Power with potentially high absolute values). However, since tree-based algorithms (e.g., XGBoost) can operate independently of scale, normalization was only required for LSTM and similar neural network models.

Finally, the dataset was split into training and test sets. The training/test split was performed at an 80–20 ratio, and reproducibility was ensured through random seeding (random state = 42). The training set is used for parameter learning of the models, while the test set is separated to evaluate the generalization performance of the models. This approach was chosen to reduce the risk of overfitting and to provide independent validation. For XGBoost, the split is performed at the observation level, whereas for the LSTM model the train–test split is performed at the country level to avoid information leakage across years of the same country, with sequences of three-year histories constructed for countries with sufficient data.

Modeling

This study adopts two different approaches for quality of life prediction: XGBoost and LSTM algorithms.

XGBoost (Extreme Gradient Boosting) is an optimized version of gradient-boosted decision trees, excelling in terms of speed, accuracy, and regularization capacity, especially for large and complex datasets. Studies have shown that XGBoost provides high predictive performance for health indicators and social welfare variables, with an advantage in error reduction compared to classical statistical modeling (Friedman, 2001, p.1189-1192; Ke, Meng & Chen, 2018, p.1).

LSTM (Long Short-Term Memory) has the power to model sequential dependencies with its special cell structure that carries information about the past. LSTM models used to predict changes in socioeconomic and environmental indicators over the years capture trends and lagged effects better (Bengio et al., 1994, p.157-160; Bai, Kolter & Koltun, 2018).

For XGBoost, we use a gradient-boosted tree model with the following specification: 500 estimators, learning rate 0.05, maximum depth 4, subsample 0.9, column subsample 0.9, L2 regularization parameter $\lambda = 1.0$, and *random_state* = 42 (implemented via XGBRegressor in Python). For the sequential model, we employ a comparatively shallow LSTM architecture to mitigate overfitting on the available panel: one LSTM layer with 64 units followed by a dense layer with 32 ReLU units and an output neuron, trained with the Adam optimizer (learning rate 0.001). Early stopping based on validation loss (patience = 20 epochs) is used for the LSTM model to avoid over-training. These choices follow standard recommendations for model regularization in small to moderate samples.

The performance of the models was evaluated on both the training and test sets using four different criteria:

- **R² (Determination Coefficient):** It shows the model's explained variance ratio, meaning how close the predictions are to the actual values.
- **MAE (Mean Absolute Error):** The average absolute difference between estimated and actual values.
- **RMSE (Root Mean Squared Error):** The square root of the mean of the squares of errors emphasizes the effect of large errors more.
- **MAPE (Mean Absolute Percentage Error):** The absolute percentage error average allows for comparing the behavior of models at different scales.

Through these performance metrics, XGBoost and LSTM models were compared; which model performs better under which conditions was empirically analyzed.

SHAP Analysis

Although machine learning models provide high accuracy, the interpretability of decision mechanisms is often limited. Therefore, in our study, the SHAP (SHapley Additive exPlanations) method was applied to increase the explainability of the XGBoost model used in quality of life prediction. SHAP is a game theory-based approach that aims to mathematically distribute the contribution of each feature to the prediction in a fair manner (Štrumbelj & Kononenko, 2014).

In this study, SHAP values were calculated and normalized for each index based on the XGBoost model, which we treat as the main explaineer due to its stability and computational efficiency for tree-based explanations. Thus, the relative effects of the eight sub-indices on quality of life could be examined from both a global (overall importance at the model level) and local (individual contributions at the country level) perspective. In the empirical section, we present both a global bar chart of mean absolute SHAP values and a SHAP summary (beeswarm) plot that displays the distribution of local contributions for each feature across all country-year observations in the test set.

SHAP-based evaluation goes beyond model performance, revealing the factors underlying the predictions. Thus, the contribution of economic, environmental, and social factors to quality of life has been assessed in a more transparent manner (Molnar, 2022, p.186-190; Petch et al., 2022, p.205).

It should be stressed that SHAP values quantify how the model's predictions change when features vary within the observed data distribution; they do not, by themselves, identify causal effects of the indices on quality of life. All interpretations in this study are therefore correlational and conditional on the model specification. When deriving policy implications from the SHAP patterns, we take this limitation into account and avoid treating feature importance as evidence that exogenous changes in a given index would mechanically produce the same magnitude of change in actual welfare outcomes.

Findings and Discussion

Robustness checks

To assess the robustness of the results and the risk of overfitting, we conducted several additional checks. First, we repeated the analysis with alternative train-test splits (70–30 and 60–40) and different random seeds; in all cases, the relative ranking of the models remained unchanged, with the LSTM model outperforming XGBoost in terms of R² and

error metrics, albeit with slightly lower absolute performance values. Second, for the LSTM model we inspected training and validation loss curves, confirming that early stopping prevents divergence between training and validation errors. Finally, we experimented with smaller and larger hidden layer sizes; using more complex architectures did not yield systematic gains, which is consistent with the limited sample size and the general recommendation to regularize recurrent models on short time series (Mostafa, 2025).

In addition, as a sensitivity check we considered time-aware splits in which earlier years were used for training and later years for testing; the qualitative ranking of models remained unchanged, although absolute performance levels decreased slightly due to the stricter validation setting.

Prediction Results

This section presents a comparative analysis of the performance of XGBoost and LSTM algorithms in predicting quality of life. The extent to which the models converge to actual values is shown in Figure 1.

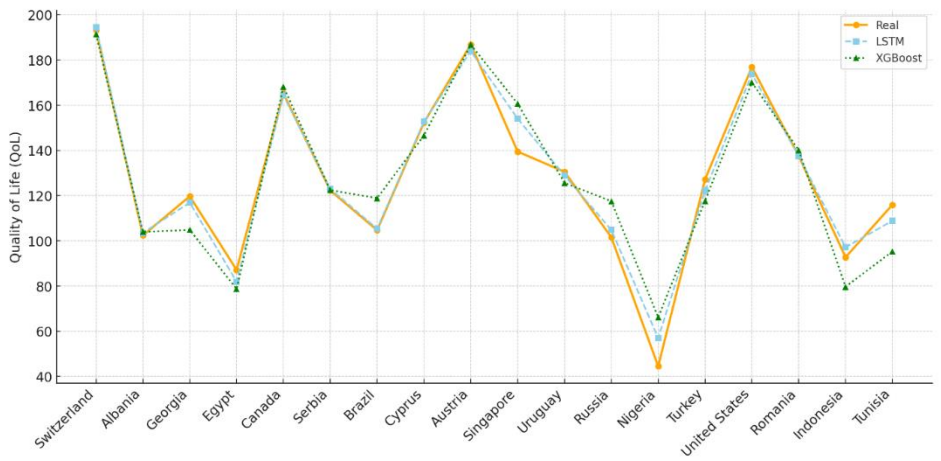


Figure 1. Comparison of XGBoost and LSTM predictions with actual quality of life values

Figure 1 presents the estimates obtained using average quality of life data for the 2016–2025 period. The blue dashed line represents the LSTM model, the green dotted line represents the XGBoost model, and the orange line represents the actual values. The graph shows that both models successfully capture the general trends, but the LSTM model provides results closer to the actual values with a lower error rate. The results of this situation are given in Table 2.

Table 2. Performance metrics of XGBoost and LSTM models

Model	R ²	MAE	RMSE	MAPE (%)
XGBoost	0.8982	9.23	11.73	9.56
LSTM	0.9781	3.75	5.44	4.06

Table 2 shows the performance metrics of the XGBoost and LSTM models on the test data. The LSTM model demonstrated superior performance compared to XGBoost, with lower error (MAE, RMSE, MAPE) and higher explanatory power (R^2) values across all metrics. In particular, the LSTM model attains a very high R^2 on the test set, indicating that it explains a large share of the variance in the quality of life variables; however, given the sample size and model complexity, these values should be interpreted with caution and in conjunction with the robustness checks and regularization strategies discussed above.

The findings show that the deep learning-based LSTM architecture is more powerful than classical gradient-boosted tree-based methods for quality of life data, where nonlinear and complex relationships are dominant. This result is consistent with studies in the literature that highlight the advantages of LSTM in capturing time dependency and interactions between variables. Furthermore, the differences in error metrics are also important from a practical standpoint: the approximate 4% MAPE value obtained with the LSTM model allows decision-makers to interpret quality of life differences between countries more reliably.

SHAP Analysis Findings

In this section, SHAP analysis was applied to increase the interpretability of the models used in quality of life estimation. The SHAP results reported below are based on the XGBoost model. SHAP values reveal the fundamental factors determining quality of life by showing the contribution and relative importance of each sub-index to the model.

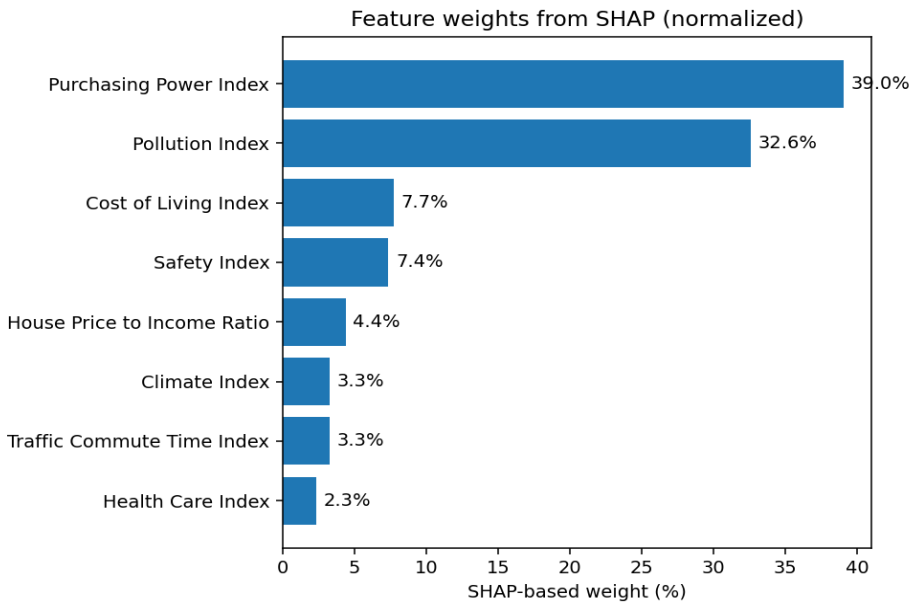


Figure 2. Relative importance of variables using SHAP analysis

As shown in Figure 2, the variables with the highest impact on quality of life estimation were the Purchasing Power Index (39%) and the Pollution Index (32.6%). This finding reveals that individuals' economic well-being and environmental conditions are decisive factors in their perception of quality of life. Other factors (Cost of Living, Safety, House

Price to Income Ratio, Climate, Traffic Commute Time, and Health Care Index) contribute to the model but have relatively lower weights. This result shows that, despite the multidimensional structure of quality of life indices, economic and environmental factors stand out.

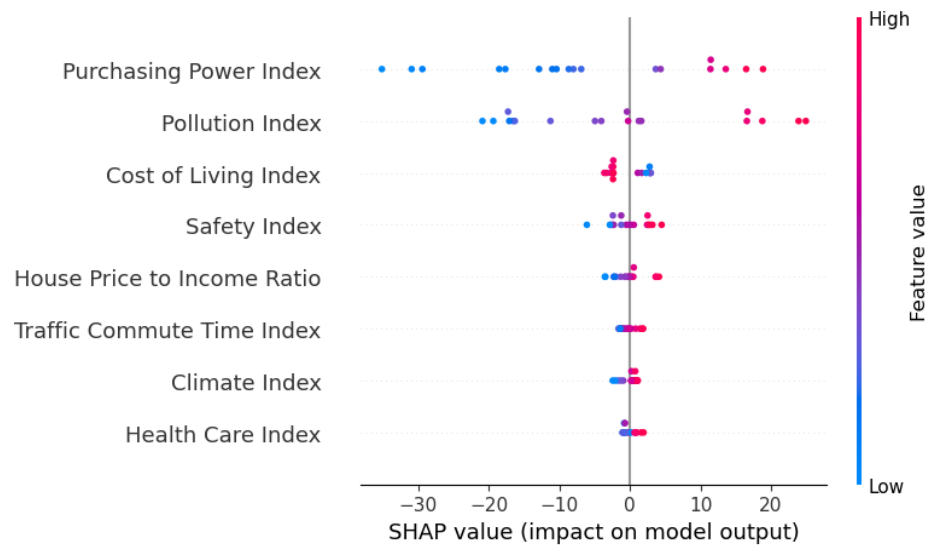


Figure 3. SHAP summary plot of feature contributions to the predicted Quality of Life Index

Figure 3 presents the SHAP summary (beeswarm) plot, which displays the distribution of local SHAP values for each feature across all country-year observations in the test set. Each dot represents a single country-year; the horizontal position shows the SHAP value (marginal contribution to the predicted QoL index), and the colour encodes the feature value (blue = low, red = high). High values of the Purchasing Power Index (red dots concentrated on the right-hand side) are associated with strongly positive SHAP values, indicating that they raise the predicted QoL. By contrast, high values of the Pollution Index (red dots clustered on the left) generate large negative SHAP values, pulling the predicted QoL down. For the remaining indices, the SHAP clouds are narrower and closer to zero, confirming that their contributions are smaller in magnitude but still non-negligible.

Taken together, Figures 2 and 3 clearly show that the most important factors affecting quality of life are economic well-being (purchasing power) and environmental quality (pollution level). High purchasing power increases individuals' capacity to meet their basic needs, while low pollution levels strengthen health and environmental sustainability conditions that directly affect quality of life.

The impact of these factors may vary between countries and over time, as suggested by the vertical spread of dots in Figure 3. For example, while high purchasing power improves quality of life in developed economies, environmental pollution can overshadow these gains in some developing countries. Similarly, factors such as healthcare or security may be more prominent in certain regions. This situation demonstrates that quality of life indicators cannot be reduced to a single dimension, and

that different priorities may emerge depending on the structural characteristics of countries.

These findings are critically important for policymakers. SHAP analysis not only identifies which factors are generally important but also enables the determination of which factors are prioritized in which countries. In this context, economic policies aimed at increasing individuals' purchasing power and environmental policies designed to reduce pollution levels are expected to contribute most directly to improving quality of life. Furthermore, factors with lower weightings, such as security, healthcare, and transportation, may also gain importance depending on the conditions of the country; therefore, country-specific analyses should not be overlooked in policy design.

Conclusion, Discussion and Suggestions

In this study, it was observed that the LSTM model demonstrated higher accuracy (higher R^2 , lower MAE/RMSE/MAPE) compared to XGBoost on the test data. Considering that comprehensive comparisons often show that ensemble methods (e.g., XGBoost) outperform deep learning on tabular data, this result particularly indicates that the time dimension (2016–2025 series) carries meaningful signals to the model and that LSTM's ability to capture sequential dependencies provides an advantage. In other words, the relative performance of LSTM is consistent with the presence of temporal patterns in the annual QoL series; however, given the size and structure of our panel, these findings should be interpreted cautiously and not as a general claim that deep networks dominate tree ensembles on tabular data (Shwartz-Ziv & Armon, 2022, p.84-90).

SHAP analyses revealed that economic and environmental dimensions came to the fore: Purchasing Power and Pollution indices are the variables that most strongly influence the model output. The literature documenting the relationship between income and life evaluation shows that high income significantly increases life evaluation, while low income leads to negative outcomes in terms of life evaluation and emotional well-being. In this context, the strong effect of purchasing power is an expected finding. Furthermore, more recent evidence shows that methodological debates on the income–well-being relationship have been reconciled with new data and methods, confirming the positive relationship between income and well-being across a wide range of incomes. These results are consistent with the predominant role of economic indicators in our model (Kahneman & Deaton, 2010, p.16489–16490).

In the environmental dimension, the negative impact of air pollution and related environmental quality indicators on subjective well-being/quality of life has been documented in numerous studies. Air pollution has been shown to reduce life satisfaction and have an additional negative effect through the health channel for different countries; the results are consistent in both individual-level panel studies and long-term data. This literature empirically supports the high negative effect of the “Pollution” index in our SHAP results (Luechinger, 2009, p.482-483).

The housing dimension (housing price/income ratio and cost of living) also enriches the discussion. Studies using microdata across Europe show that the housing cost burden reduces subjective well-being and life satisfaction; which explains why housing affordability is a critical channel in quality of life and why the “Housing Price/Income Ratio” and “Cost of Living” remain meaningful in our model (Acolin & Reina, 2022, 1789).

Finally, the differences between countries are noteworthy. While purchasing power is decisive in some countries, the impact of pollution or cost pressures is more severe in others. Classic studies on the conceptual and measurement framework of urban/environmental quality and human well-being emphasize that elements such as indicator selection, scale, time horizon, and context (e.g., urban morphology, climate, infrastructure, institutions) can significantly shape the findings; this indicates that a one-size-fits-all policy prescription cannot be adapted to every country (Pacione, 2003, p.23-24).

This study has some limitations. First, since the data used in the analysis consists of annual average values, seasonal fluctuations, short-term changes, and sudden shocks cannot be captured. Second, the study only covers countries for which data is available as of 2025. This limits the generalizability of the model due to data gaps or incomplete series in some countries. Third, while SHAP analysis shows the overall contribution of variables to the model, local conditions, cultural differences, and country-specific events may not always be fully reflected in the models. Finally, the data sources used (e.g., user-contributed indices such as Numbeo) involve perception-based inquiries, which may lead to representation issues and uncertainties regarding data consistency in some countries.

In particular, Numbeo relies on voluntary user contributions, which can generate self-selection bias towards more digitally connected and urban populations, unequal sample sizes across countries and years, and potential framing effects in survey questions. These issues may introduce measurement error in both the sub-indices and the composite QoL index, and may attenuate or distort some of the relationships captured by the machine-learning models. As a result, the findings should be interpreted as conditional on the Numbeo sampling frame rather than as fully representative of the entire resident population in each country.

In addition, because the composite Quality of Life Index is mechanically constructed from the same sub-indices that are used as predictors in our models, the estimated relationships are inherently constrained by this index definition. As discussed in the methodological section, part of the high predictive performance may therefore reflect the reconstruction of Numbeo's aggregation rule rather than the prediction of an independent welfare outcome. Future research could address this limitation by either using external well-being measures (e.g. life satisfaction, health-related quality of life, life expectancy) as targets or by augmenting the feature set with socio-economic indicators that are not directly embedded in the QoL formula.

In this study, XGBoost and LSTM algorithms were compared using quality of life data from the 2016–2025 period, and SHAP analysis was used to identify the most important factors determining quality of life. The findings show that both models successfully capture quality of life trends, but the LSTM model outperforms XGBoost with higher accuracy ($R^2=0.9781$) and lower error (MAE, RMSE, MAPE) values. However, given the limited sample size and the definitional link between the composite index and its components, this high level of performance should be interpreted with caution and in light of the robustness checks discussed above. SHAP analysis revealed that Purchasing Power and Pollution indices are particularly decisive in predicting quality of life, while other factors carry different levels of importance depending on the country context.

The findings clearly indicate that quality of life policies should focus on increasing economic prosperity and reducing environmental pollution. Economic policies that increase purchasing power (minimum wage regulation, tax breaks, employment policies) are expected to contribute to quality of life. Similarly, environmental regulations aimed at reducing air pollution and sustainable energy policies are expected to strengthen individuals' perception of health and well-being. However, in some countries, housing prices, cost of living, security, or health services may be priorities; therefore, it is important to develop country-specific customized policies.

The results of the study also provide a field of application for decision-makers. In particular, national and local governments can use the priority ranking provided by the SHAP analysis to determine policy priorities and direct limited resources to areas with the highest impact.

The results of this study also offer important insights for future research:

- Using data at higher frequencies (e.g., quarterly or monthly) may enable analysis of the effects of short-term fluctuations on quality of life.
- Panel data models and multilevel approaches may contribute to a better understanding of cross-country differences.
- Comparing other explainable AI methods (e.g., LIME, PDP, ALE) besides SHAP may provide policymakers with a richer set of insights.
- The performance of different deep learning architectures (e.g., GRU, Transformer-based models) in QoL predictions can be examined and compared with LSTM.
- Testing geographically weighted models or graph neural networks (GNNs) to capture spatial dependencies may more accurately capture the effect of regional interactions on QoL.

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